ASADULLAH ALEEM

Software Engineer | Python | FastAPI | React | SQL

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PROFESSIONAL SUMMARY

Software Engineering graduate (BS, COMSATS University Lahore, 2026) with i hands-on project experience in full-stack and backend development using Python, FastAPI, React, Next.js, and PostgreSQL. Builds RESTful APIs, database-driven applications, and role-based access control (RBAC) systems across frontend, backend, and database layers, with applied AI/ML experience. Published research on security and reliability risks in healthcare software.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Dart, HTML, CSS
Backend & APIs: FastAPI, RESTful APIs, Node.js, Microservices Architecture
Frontend: React.js, Next.js, Flutter
Databases: PostgreSQL, MongoDB, Supabase
Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, Database Fundamentals, Role-Based Access Control (RBAC), Secure Application Design
AI/ML: Machine Learning, SVM, KNN, Random Forest, OpenCV
Tools: Git, GitHub

PROFESSIONAL EXPERIENCE

Full-Stack Dev Engineering Intern | NETSOL Technologies Lahore, Pakistan | Jul – Sep 2026

- Develop and test RESTful API endpoints in Python and FastAPI within a microservices architecture, integrating PostgreSQL to support core backend workflows.
- Build and refine React front-end components alongside senior engineers, translating product requirements into tested UI features.
- Follow a Git-based code review and deployment workflow to deliver production-ready code across the stack.

SELECTED PROJECTS

Diabetes Prediction & Risk Mitigation Platform — Full-Stack + Applied ML Published Research

- Designed and built a full-stack healthcare platform with Next.js (web), Flutter (mobile), and Supabase/PostgreSQL, unifying patient, doctor, and admin workflows behind one secure backend.
- Implemented role-based access control (RBAC), appointment booking, real-time video consultations, lab-report management, and real-time notifications.
- Built and compared SVM, KNN, and Random Forest models for diabetes risk prediction, integrating results into the application through REST APIs.

Auto Parts Inventory & Sales Management System

- Designed a role-based inventory system to manage parts, suppliers, and sales records, replacing manual tracking with structured, access-controlled workflows.
- Implemented role-based authorization with separate permission tiers for admin, staff, and sales users to protect data integrity across the sales process.

Code Smell Detection Tool — Python Static Analysis

- Developed a Python static-analysis tool that scans source code to detect code smells and maintainability risks, supporting automated code-quality checks.

Number Plate Detection System — Computer Vision | Python

- Built a Python pipeline using OpenCV and machine learning to detect and read vehicle license plates from images, removing manual plate lookup.

Bird Species Detection System — Machine Learning / Computer Vision

- Trained an image classification model to distinguish multiple bird species, applying data preprocessing and model evaluation.

EDUCATION

BS Software Engineering — COMSATS University Lahore | CGPA: 3.28/4.0 2022 – 2026

CERTIFICATIONS

- The Ultimate React Course 2025 — Udemy
- Python Fundamentals for Beginners — Great Learning Academy

PUBLICATION

A. Aleem, "Risk Identification and Mitigation for Diabetes Prediction and Control System in Healthcare Software," SSRG International Journal of Computer Science and Engineering, Vol. 12, Issue 5, pp. 19–28, 2025. DOI: 10.14445/23488387/IJCSEV12I5P103